

Problem Set 9

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1 Question 7

When comparing how many additional X variables there were between the original housing data and the prepped data, the numbers are 14 and 75, respectively.

2 Question 8

Optimal λ is 0.00139, In sample RSME 0.0677, Out of sample RSME 0.170

3 Question 9

Optimal λ is 0.0233, In sample RSME 0.0682, Out of sample RSME 0.173

4 Question 10

You would not be able to estimate a simple linear regression when you have more columns than rows because there would be infinitely many possible solutions.

Both models have a much lower in-sample RMSE (0.068) than out-of-sample RMSE (0.17), which means they fit the training data better than new data. This is a sign of high variance (overfitting). Regularization helped, but the gap suggests the models are still closer to the high-variance end of the tradeoff.